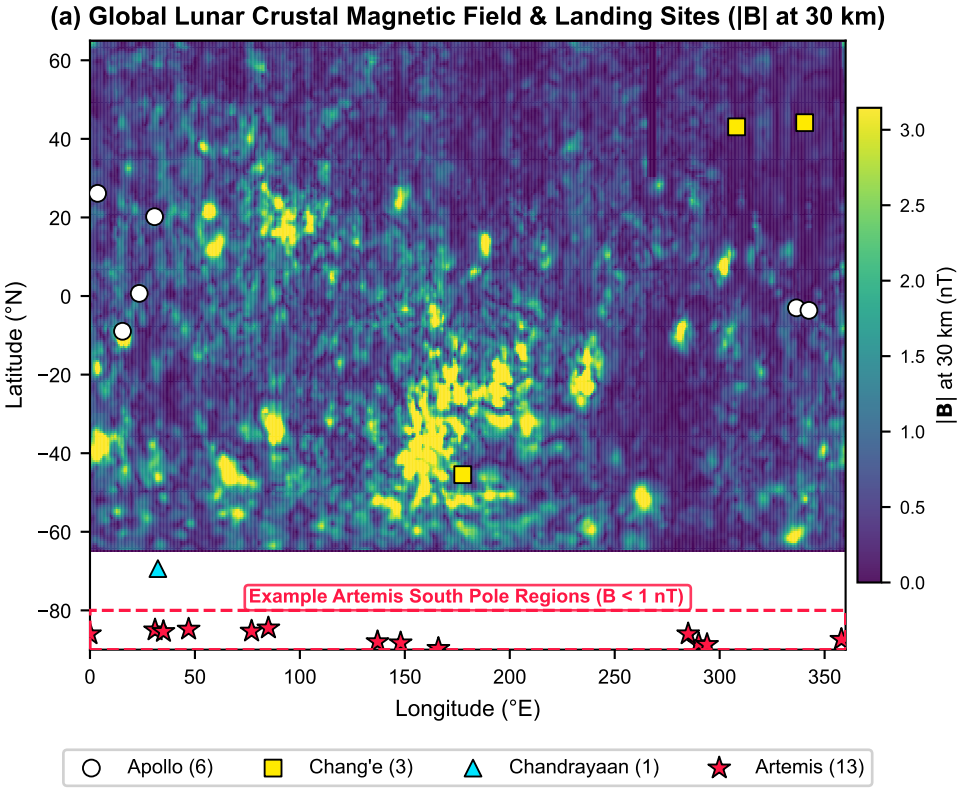
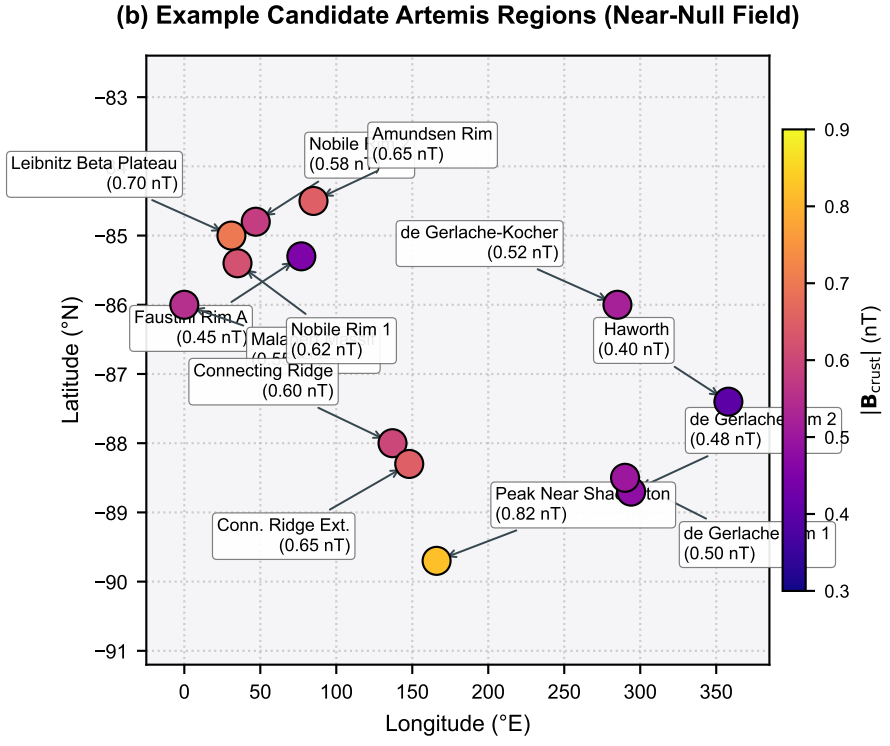
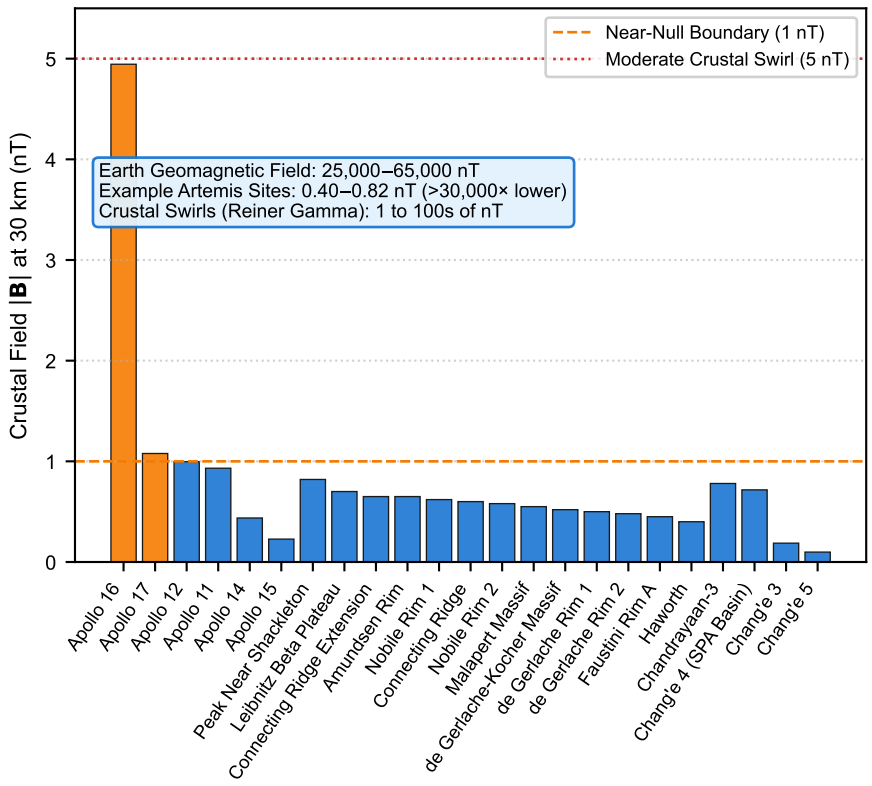






(d) Field Superposition at the Biological Volume

$$\mathbf{B}_{total}(t, \mathbf{x}) = \mathbf{B}_{crust} + \mathbf{B}_{ext} + \mathbf{B}_{lander} + \mathbf{B}_{payload} + \mathbf{B}_{control}$$

Net Vector Sum Governs Primary Spin-Chemical Biomarkers

Lunar Plant Growth Enclosure (e.g., LEAF / BLSS)

$\mathbf{B}_{crust}$	**[0.4 – 0.8 nT]**	Lunar South Pole paleomagnetic void
$\mathbf{B}_{ext}$	**[1 – 5 nT]**	Solar wind IMF / magnetotail shadow
$\mathbf{B}_{lander}$	**[50 – 500 μT]**	Lander avionics, DC power buses & batteries [Confounder]
$\mathbf{B}_{payload}$	**[15 – 20 nT]**	Payload fans, fluidic pumps, heaters, LEDs [Ripple]
$\mathbf{B}_{control}$	**[40 – 50 μT]**	Active 3-axis compensation / causal rescue coils

CRITICAL: LANDER AVIONICS ARTIFACT RISK

Unshielded DC currents (50–500 μT) follow a steep $1/r^3$ falloff and can swamp the ambient null field. Sample-level 3-axis canopy magnetometry (<1 W, 50 g) is essential to decouple true lunar hypomagnetic effects from avionics noise.